Name : Piyush Gupta

Batch : T12

Roll no. : 2401033

Branch : AI&DS

Experiment-05

AIM: To understand Continuous integration, Jenkins

installation. LO mapped: LO3

Theory:

Continuous Integration and Jenkins

Continuous Integration (CI) is a development practice where code is frequently integrated, built, and tested to detect errors early and improve software quality.

Jenkins is an open-source automation server used for implementing CI/CD. It automates tasks such as building, testing, and deploying software and supports integration with tools like Git, Maven, Docker, Kubernetes, and AWS.

Key Functions:

- Automates build, testing, and deployment.
- Supports numerous plugins and development tools.
- Triggers jobs through webhooks, schedules, or manual execution.

Jenkins Architecture

1. Jenkins Controller (Master)

- Manages jobs and configurations.
- Schedules builds and assigns them to agents.
- Provides the Jenkins Web UI.

2. Jenkins Agent (Node)

- Executes build, test, and deployment tasks.
- Allows distributed builds across multiple machines.

Types of Jenkins Pipelines

A Jenkins Pipeline automates the complete CI/CD workflow using a **Jenkinsfile**.

A. Declarative Pipeline

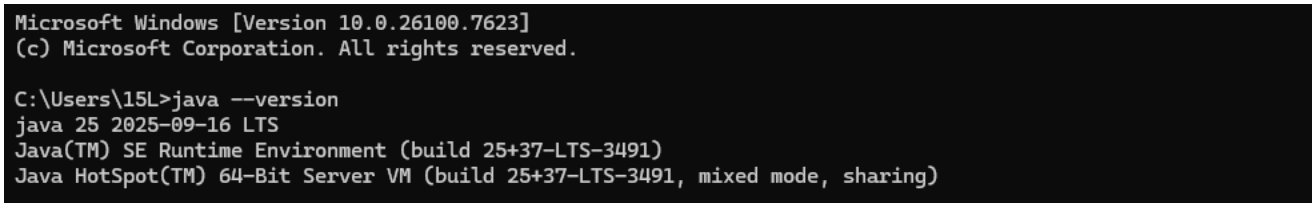
- Written inside the pipeline {} block.
- Simple, structured, and easy to read.
- Provides built-in syntax validation.
- Uses sections like agent, stages, stage, and steps.
- Best for standard CI/CD workflows.

B. Scripted Pipeline

- Written inside the node {} block.
- Uses Groovy scripting language.
- Highly flexible and customizable.
- Suitable for complex workflows with conditions and loops.
- More difficult to learn than Declarative Pipeline.

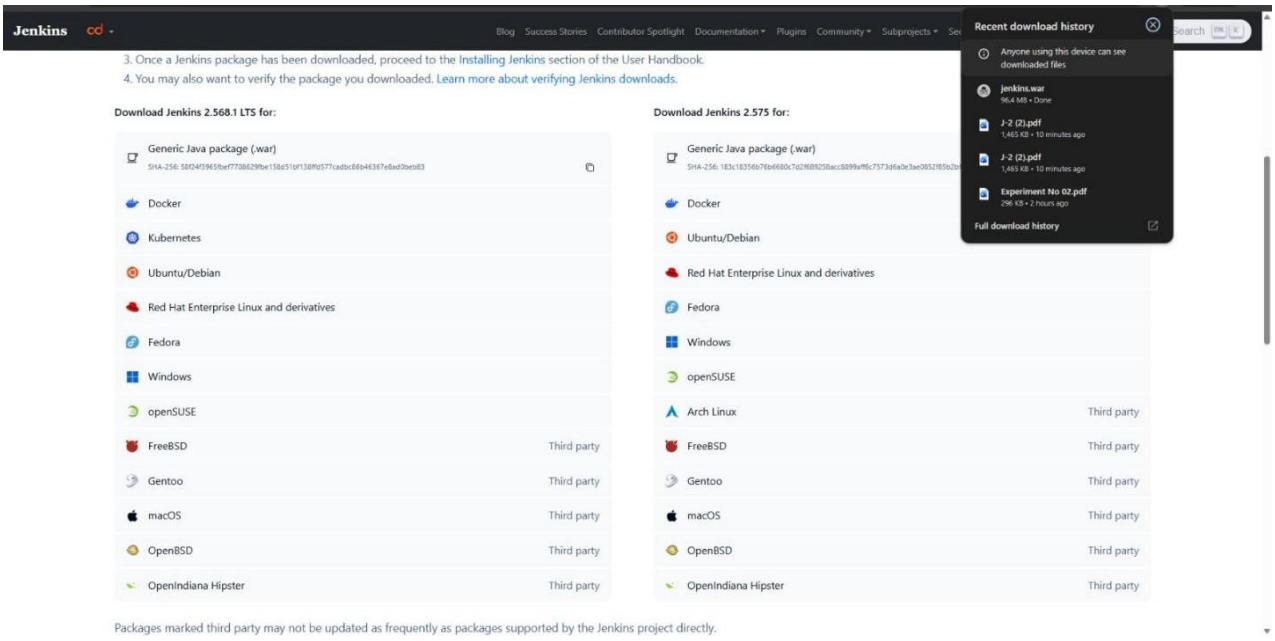
OUTPUT:

Step 1: Java (JDK) Installation on Verification



Step 2: Jenkins Server Launch

Downloaded the Generic Java package (jenkins.war) from the official website.

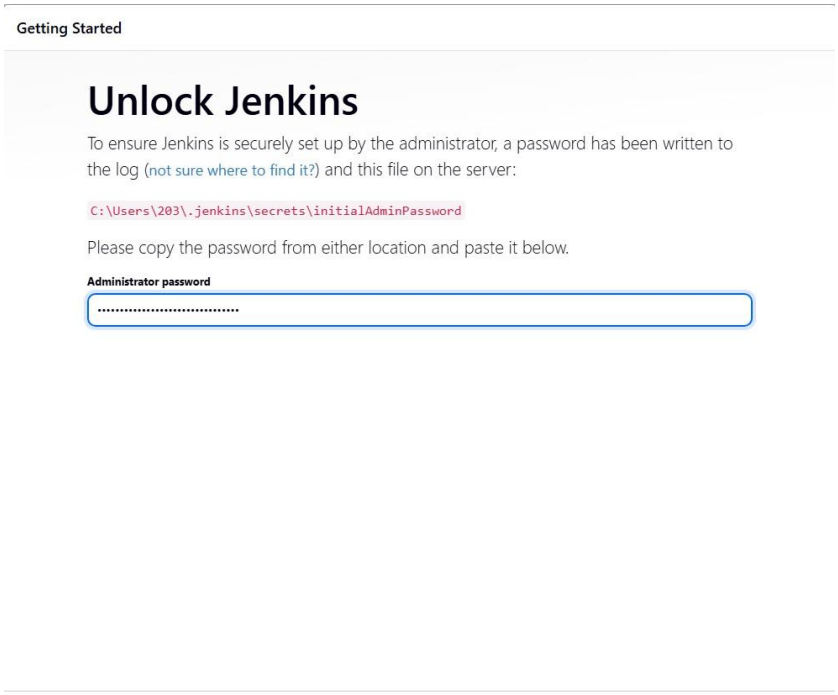


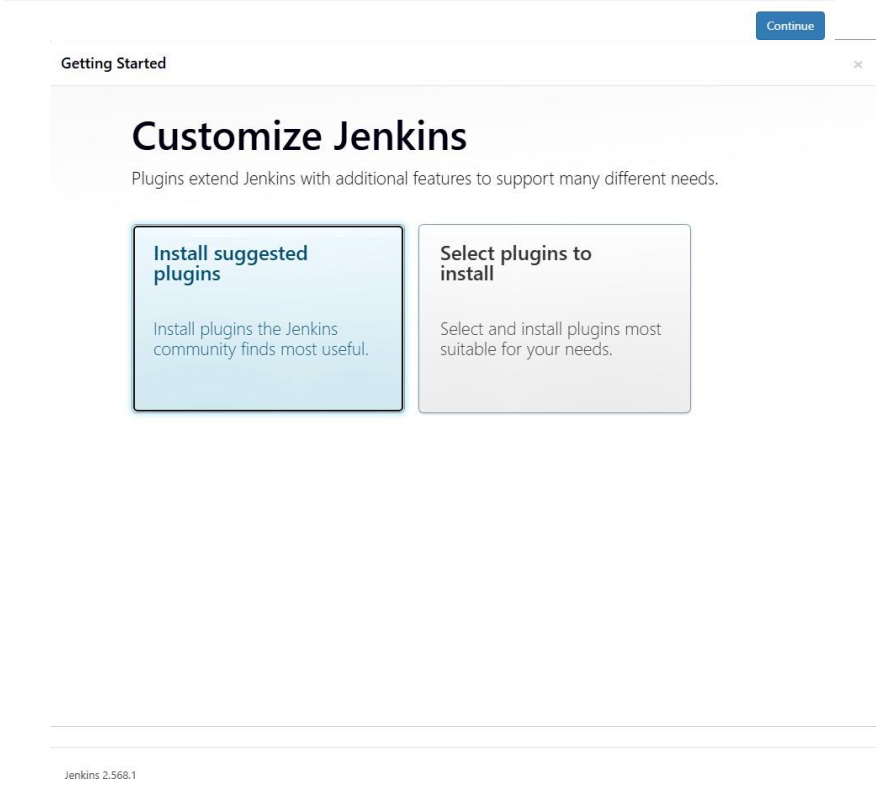
Executed Jenkins using the --enable-future-java flag to support JDK 22:

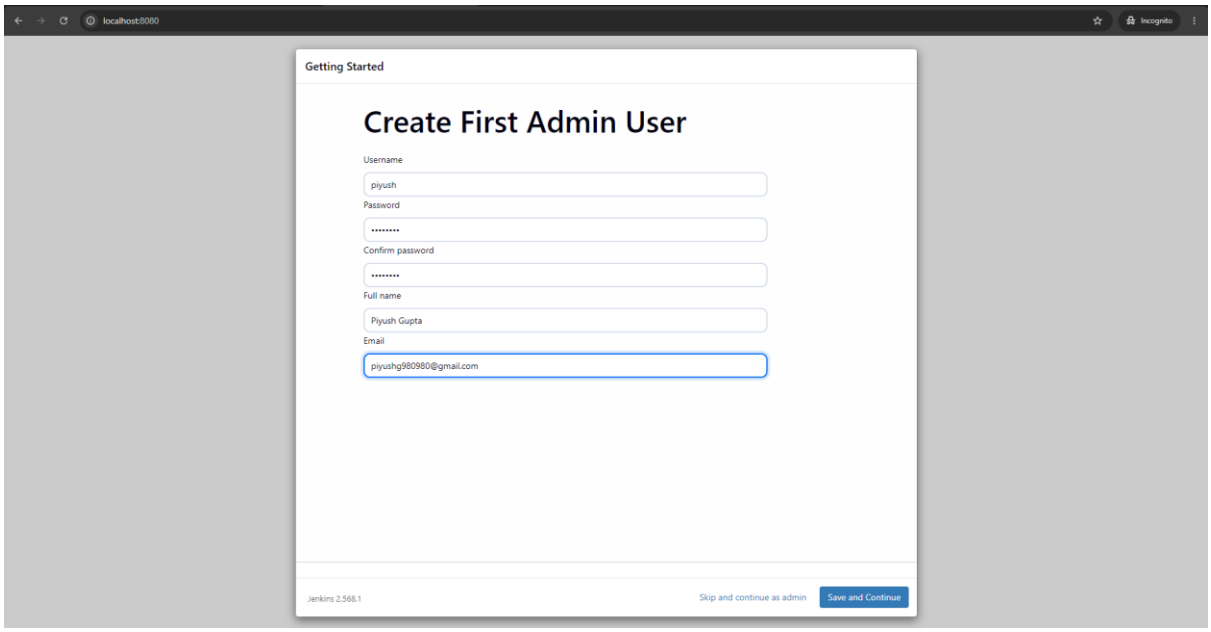


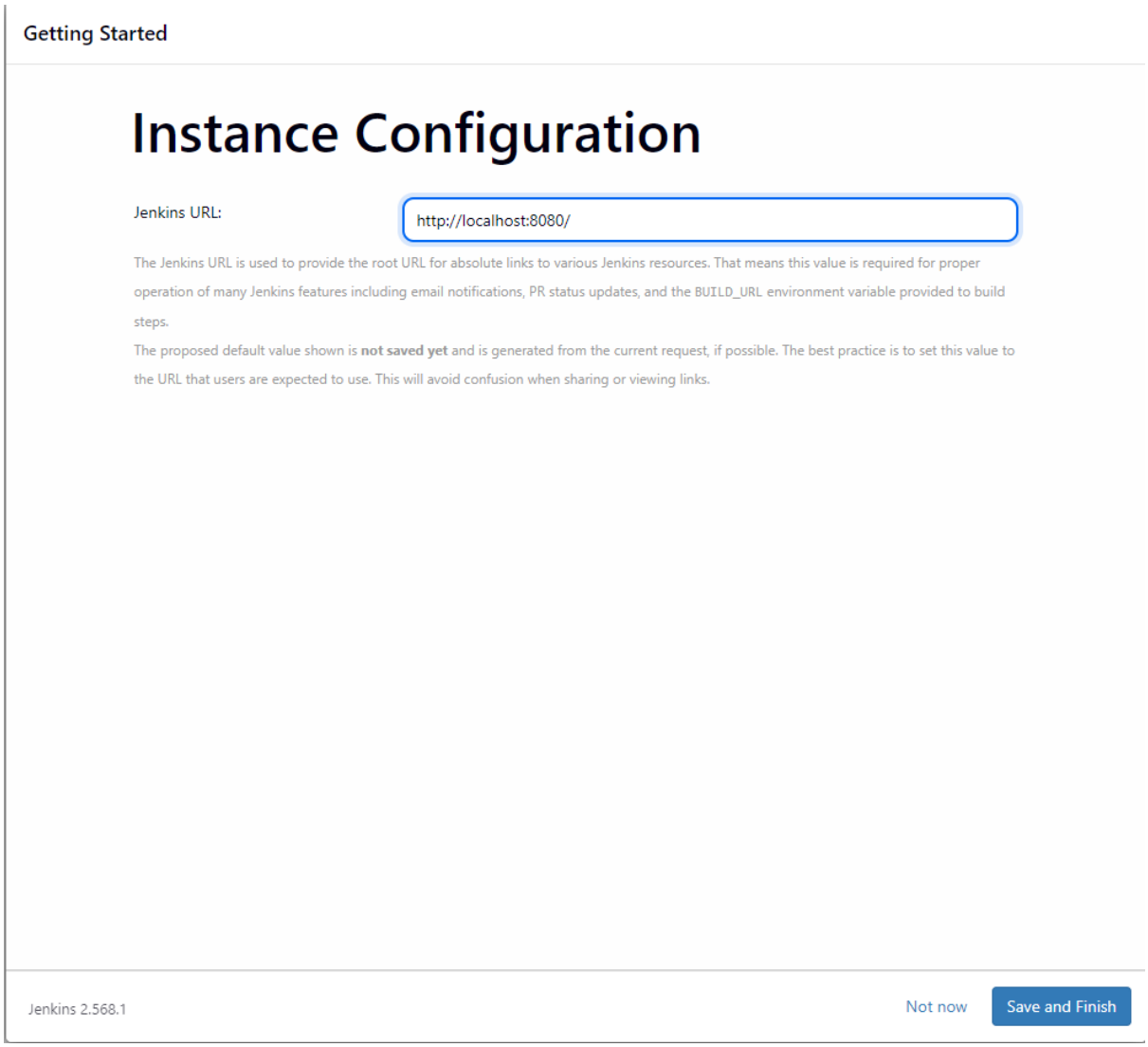
```
2026-08-04 05:13:40.354+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: trilead-api
2026-08-04 05:13:40.354+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of ssh-slaves on behalf of admin
2026-08-04 05:13:40.355+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading ssh-slaves
2026-08-04 05:13:41.693+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: ssh-slaves
2026-08-04 05:13:41.693+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of matrix-auth on behalf of admin
2026-08-04 05:13:41.696+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading matrix-auth
2026-08-04 05:13:43.242+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: matrix-auth
2026-08-04 05:13:43.242+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of ldap on behalf of admin
2026-08-04 05:13:43.242+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading ldap
2026-08-04 05:13:44.850+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: ldap
2026-08-04 05:13:44.850+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of jsoup on behalf of admin
2026-08-04 05:13:44.851+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading jsoup
2026-08-04 05:13:46.291+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: jsoup
2026-08-04 05:13:46.291+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of email-ext on behalf of admin
2026-08-04 05:13:46.293+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading email-ext
2026-08-04 05:13:47.811+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: email-ext
2026-08-04 05:13:47.811+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of mailer on behalf of admin
2026-08-04 05:13:47.811+0000 [id=165]   INFO    h.m.UpdateCenter$InstallationJob#_run: Skipping duplicate install of: Mailer@534.v1b_36f5864073
2026-08-04 05:13:47.812+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: mailer
2026-08-04 05:13:47.812+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of theme-manager on behalf of admin
2026-08-04 05:13:47.812+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading theme-manager
2026-08-04 05:13:49.352+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: theme-manager
2026-08-04 05:13:49.352+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Starting the installation of dark-theme on behalf of admin
2026-08-04 05:13:49.353+0000 [id=165]   INFO    h.m.UpdateCenter$UpdateCenterConfiguration#download: Downloading dark-theme
2026-08-04 05:13:51.807+0000 [id=165]   INFO    h.model.UpdateCenter$DownloadJob#run: Installation successful: dark-theme
2026-08-04 05:13:51.807+0000 [id=165]   INFO    h.m.UpdateCenter$CompleteBatchJob#run: Completing installing of plugin batch?
2026-08-04 05:13:53.379+0000 [id=165]   INFO    h.p.b.g.GlobalTimeOutConfiguration#load: global timeout not set
2026-08-04 05:13:55.187+0000 [id=165]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Ant.AntInstaller
2026-08-04 05:13:57.391+0000 [id=165]   INFO    h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.plugins.gradle.GradleInstaller
2026-08-04 05:13:57.804+0000 [id=320]   INFO    jenkins.InitReactorRunner$1#onAttained: Started initialization
2026-08-04 05:13:57.806+0000 [id=333]   INFO    jenkins.InitReactorRunner$1#onAttained: Listed all plugins
2026-08-04 05:13:57.807+0000 [id=338]   INFO    jenkins.InitReactorRunner$1#onAttained: Prepared all plugins
2026-08-04 05:13:57.823+0000 [id=347]   INFO    jenkins.InitReactorRunner$1#onAttained: Started all plugins
2026-08-04 05:13:57.826+0000 [id=353]   INFO    jenkins.InitReactorRunner$1#onAttained: Augmented all extensions
2026-08-04 05:13:57.826+0000 [id=325]   INFO    jenkins.InitReactorRunner$1#onAttained: System config loaded
2026-08-04 05:13:57.827+0000 [id=334]   INFO    jenkins.InitReactorRunner$1#onAttained: System config adapted
2026-08-04 05:13:57.833+0000 [id=356]   INFO    jenkins.InitReactorRunner$1#onAttained: Loaded all jobs
2026-08-04 05:13:57.833+0000 [id=346]   INFO    jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
2026-08-04 05:13:58.067+0000 [id=352]   INFO    jenkins.InitReactorRunner$1#onAttained: Completed initialization
2026-08-04 05:13:58.069+0000 [id=165]   INFO    h.m.UpdateCenter$CompleteBatchJob#run: Completed installation of 92 plugins in 1 min 48 sec
```

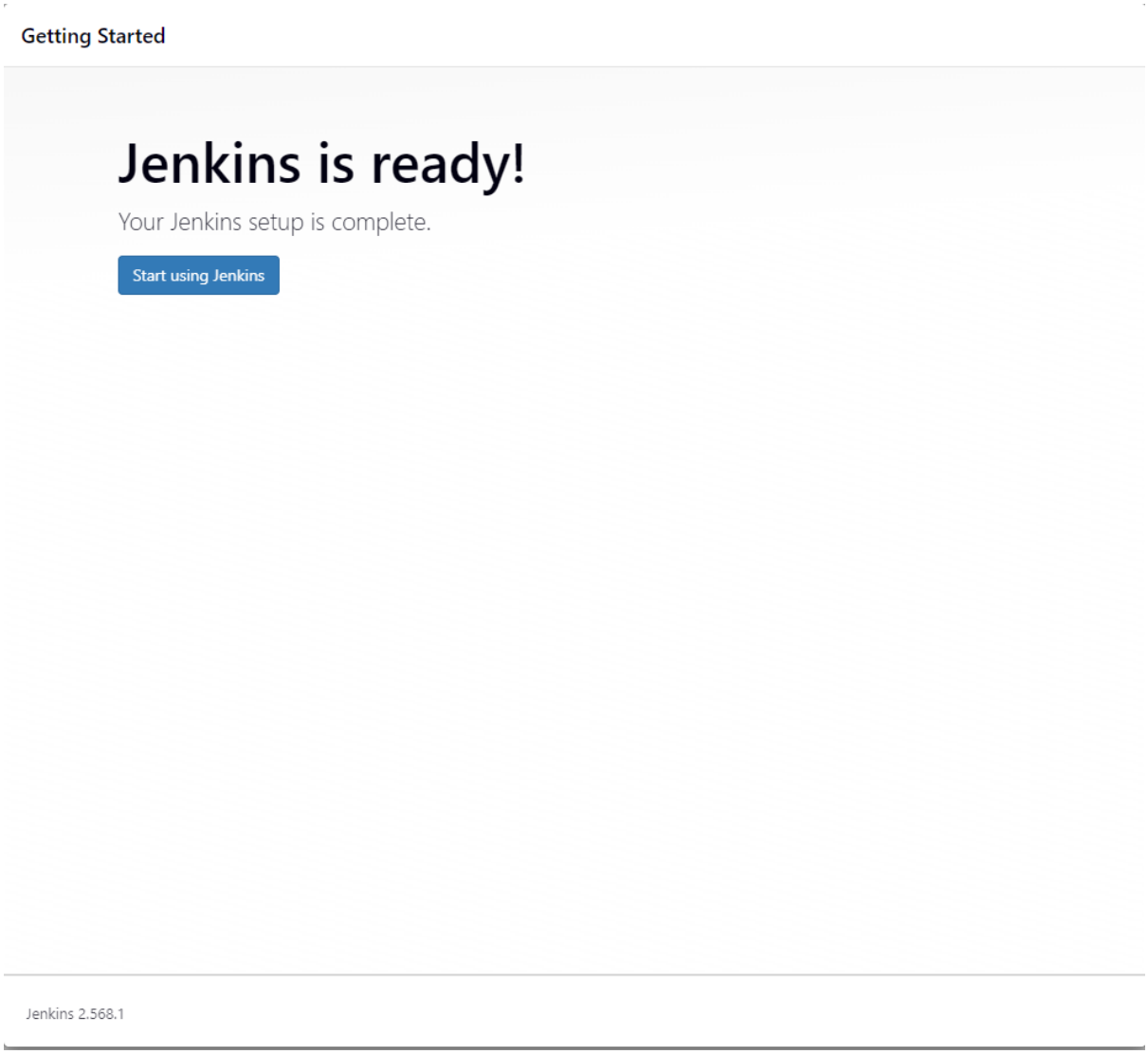
Step 3: Web Setup & Admin Configuration



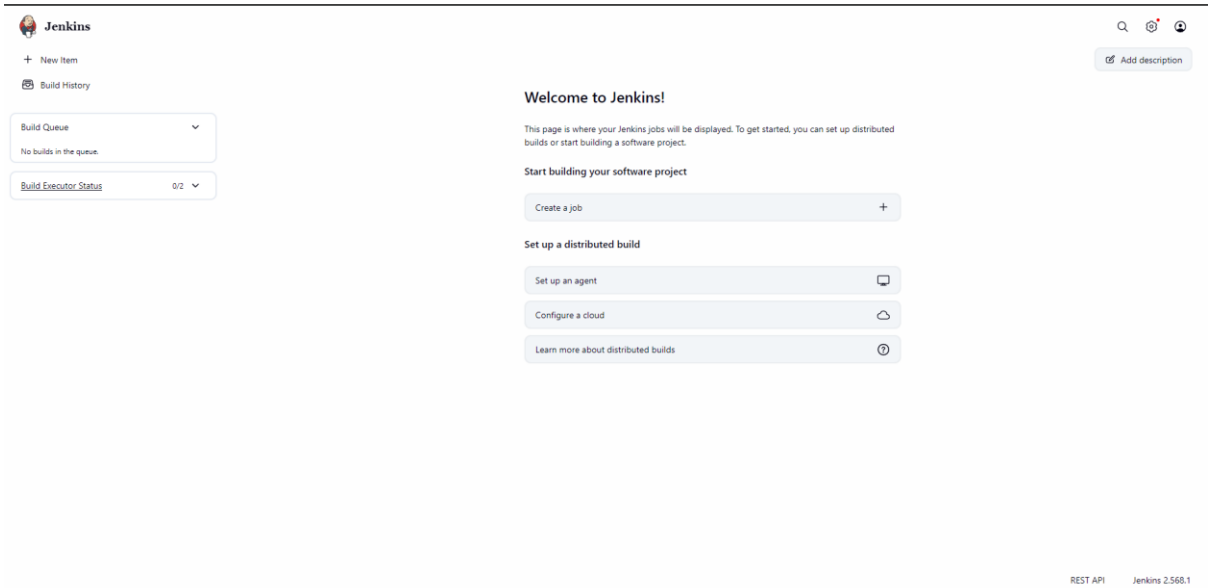


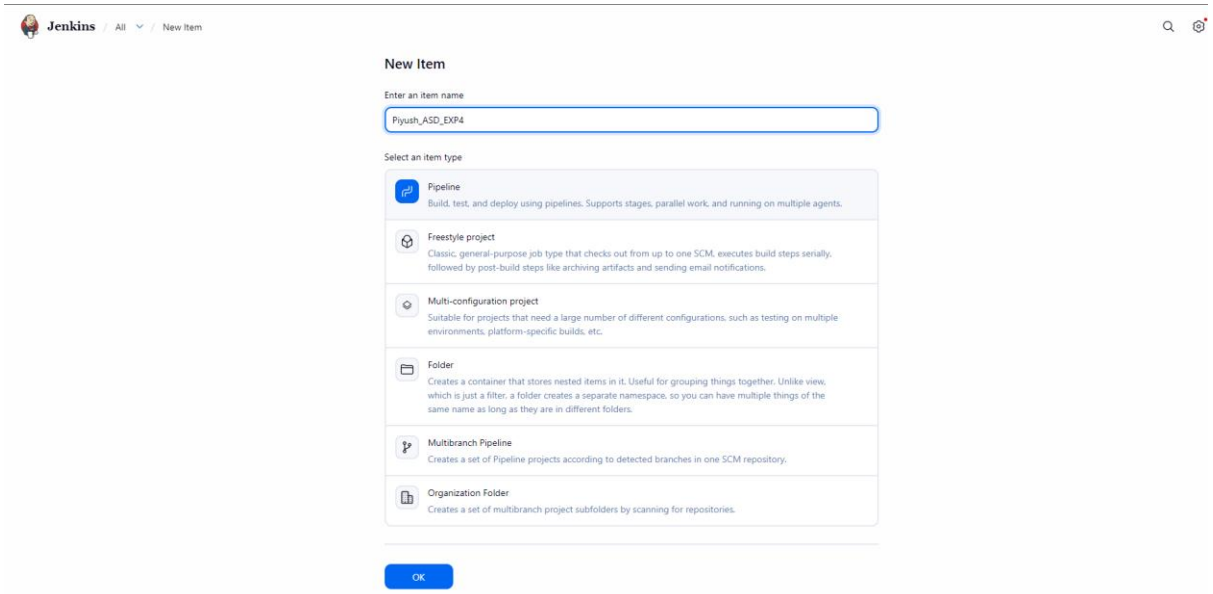

localhost:8080
Incognito
Getting Started
Create First Admin User
Username
piyush
Password
Confirm password
Full name
Piyush Gupta
Email
piyushg980980@gmail.com
Jenkins 2.568.1
Skip and continue as admin
Save and Continue


Getting Started
Instance Configuration
Jenkins URL:
http://localhost:8080/
The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.
The proposed default value shown is not saved yet and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.
Jenkins 2.568.1
Not now
Save and Finish



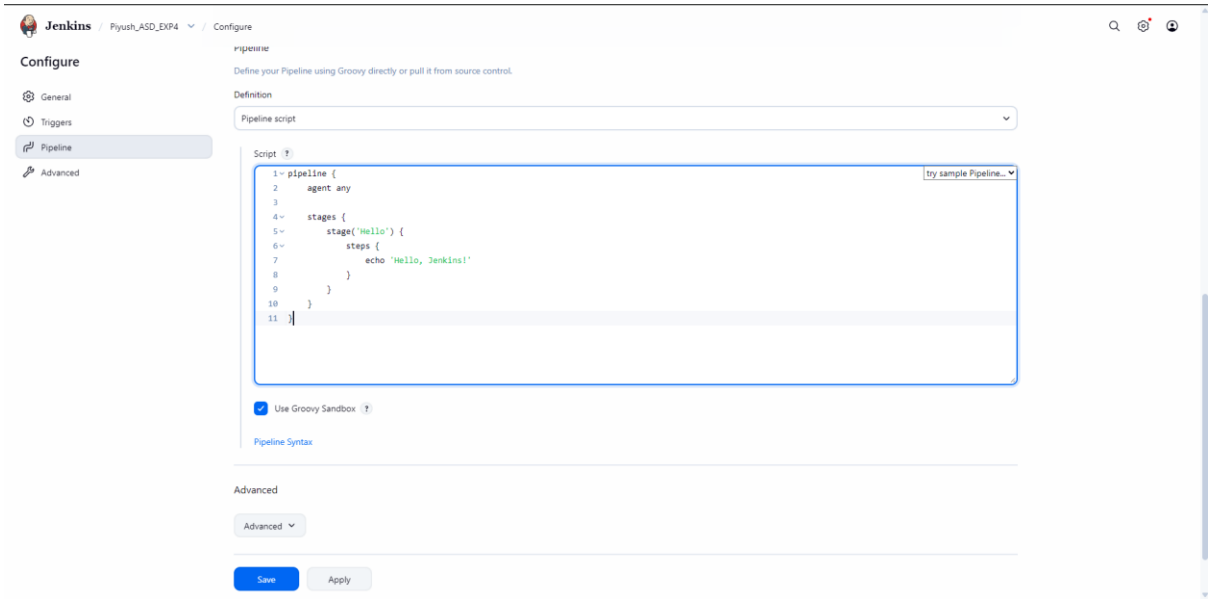
Step 4: Creating Pipelines



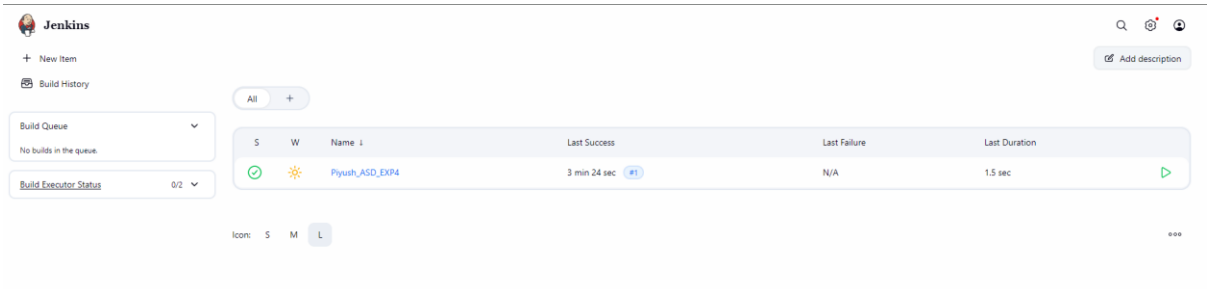

Jenkins
All
New Item
New Item
Enter an item name
Piyush_ASD_EXP4
Select an item type
Pipeline
Build, test, and deploy using pipelines. Supports stages, parallel work, and running on multiple agents.
Freestyle project
Classic, general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.
Multi-configuration project
Suitable for projects that need a large number of different configurations, such as testing on multiple environments, platform-specific builds, etc.
Folder
Creates a container that stores nested items in it. Useful for grouping things together. Unlike view, which is just a filter, a folder creates a separate namespace, so you can have multiple things of the same name as long as they are in different folders.
Multibranch Pipeline
Creates a set of Pipeline projects according to detected branches in one SCM repository.
Organization Folder
Creates a set of multibranch project subfolders by scanning for repositories.
OK

Code:

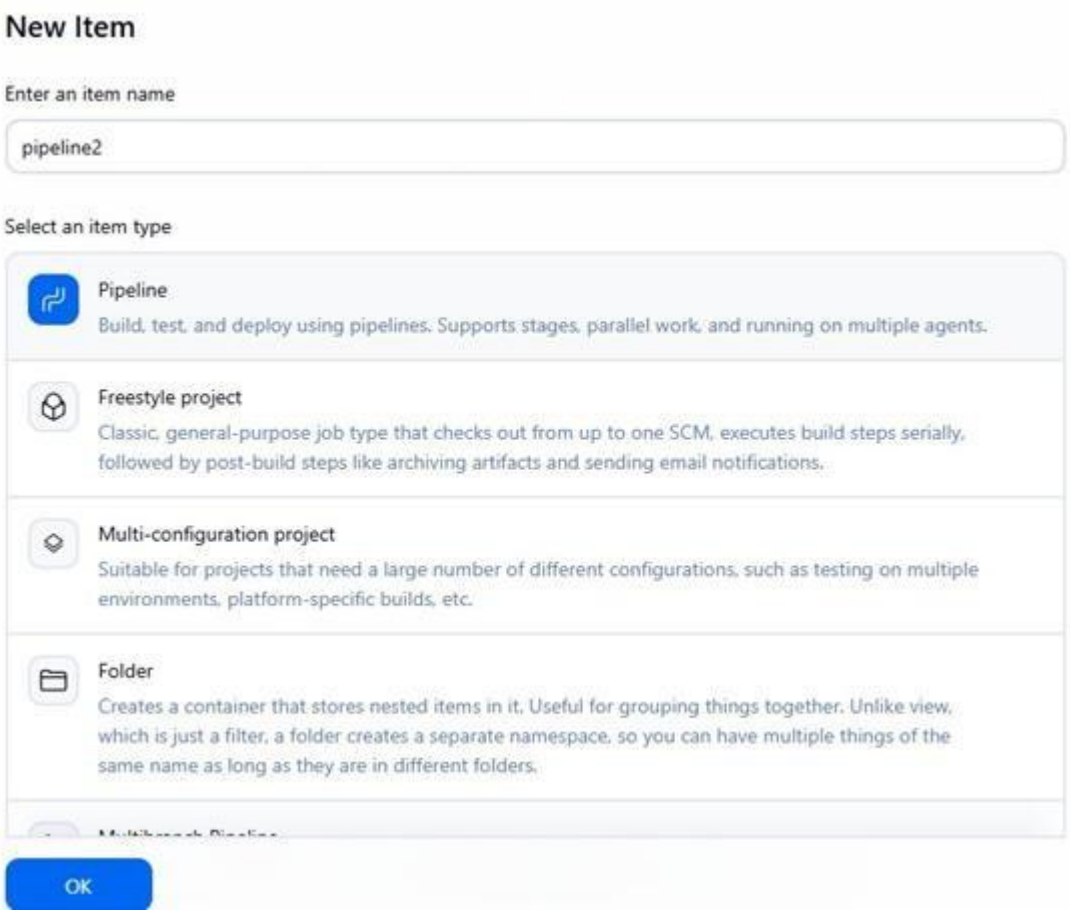
```
pipeline {
        agent
        any
        stages {
                stage('Hello') {
                        steps {
                                echo 'Hello, Jenkins!'
                        }
                }
        }
}
```




Jenkins / Piyush_ASD_EXP4 / #1
Status
Changes
Console Output
Delete build '#1'
Pipeline Overview
Edit build information
Restart from Stage
Replay
Pipeline Steps
Workspaces
Console
Started by user Piyush Gupta
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in C:\Users\15L\.jenkins\workspace\Piyush_ASD_EXP4
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Hello)
[Pipeline] echo
Hello, Jenkins!
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS


Jenkins
New Item
Build History
Build Queue
No builds in the queue.
Build Executor Status
0/2
All
Add description
S
W
Name
Last Success
Last Failure
Last Duration
Piyush_ASD_EXP4
3 min 24 sec
#1
N/A
1.5 sec
Icon: S M L

Step 5: Creating Scripted Pipeline



Code:

```
node
{
    stage('Hello') {
        echo 'Hello, Scripted Jenkins Pipeline!'
    }
}
```



CONCLUSION: LO1, LO3

In this experiment, we learned how to create Jenkins Pipelines to automate the build, test, and deployment process. We explored both Declarative and Scripted pipelines, understanding their structure, benefits, and use cases. By implementing different pipeline examples, we demonstrated how Jenkins enables efficient, repeatable, and reliable CI/CD workflows.